

Indian Institute of Technology Roorkee
Mid-Term Examination
ECN 205: Analog Circuits
Course Instructors: Sourajeet Roy and Sudeb Dasgupta

Duration: 1.5 hours

MM:40

Note: Attempt all questions. Assumptions taken may be stated.

Q1. For the circuit given in Fig. 1 below assume all $\lambda = 0$

- (i) Will the presence of R_{D1} reduce or increase the transconductance $G_m = i_{out}/v_{in}$ when $v_{out} = 0$ of the amplifier? Justify your answer. **(3 marks)**
- (ii) What is the transconductance of the circuit G_m ? Can you support your answer without writing any equations? **(5 marks)**
- (iii) Determine the voltage gain of the circuit. **(5 marks)**
- (iii) If $R_{D1} = \infty$, the amplifier is of the _____ topology. **(2 marks)**

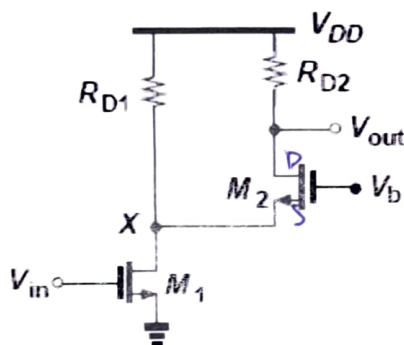


Fig. 1

Q2. Consider the differential amplifier circuit shown below. Assume $\lambda = 0$ for $M_{1,2}$ but not for $M_{3,4}$. Let the tail current sources on the left- and right-hand side be equal.

- (a) Calculate the differential model gain. **(6 marks)**
- (b) Calculate the common mode to differential mode conversion gain assuming that the mismatch lies in the r_{o3} and r_{o4} resistances (i.e., $r_{o3} = r_{o4} + \Delta r$) **(6 marks)**

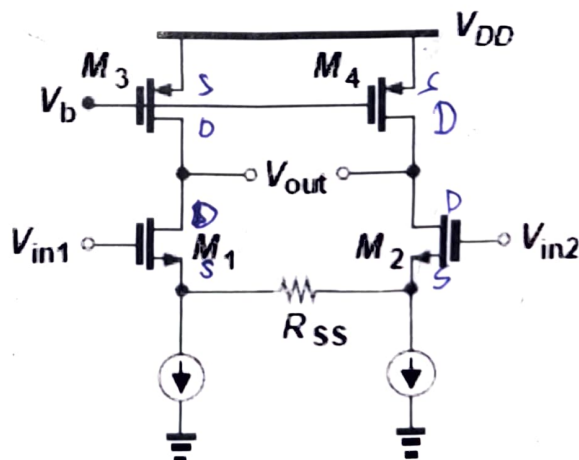


Fig. 2

Q3. Design the multi-stage amplifier shown below, such that $I_{D1} = 0.2 \text{ mA}$, $I_{D2} = 0.5 \text{ mA}$, $V_{DS1} = V_{DS2} = 6V$ and $R_{in} = 100 \text{ k}\Omega$. Let $R_{S1} = 10 \text{ k}\Omega$.
 Given: $K'_{n1} \left(\frac{W}{L}\right)_1 = 500 \mu\text{A}/V^2$, $K'_{n2} \left(\frac{W}{L}\right)_2 = 200 \mu\text{A}/V^2$, $V_{TN1} = V_{TN2} = 1.2 \text{ V}$, and neglect channel length modulation. Also, $K'_{n1} = \mu_{n1} C_{ox1}$, $K'_{n2} = \mu_{n2} C_{ox2}$. Assume all capacitors to be short circuits in small signal domain and open circuits in large signal domain.

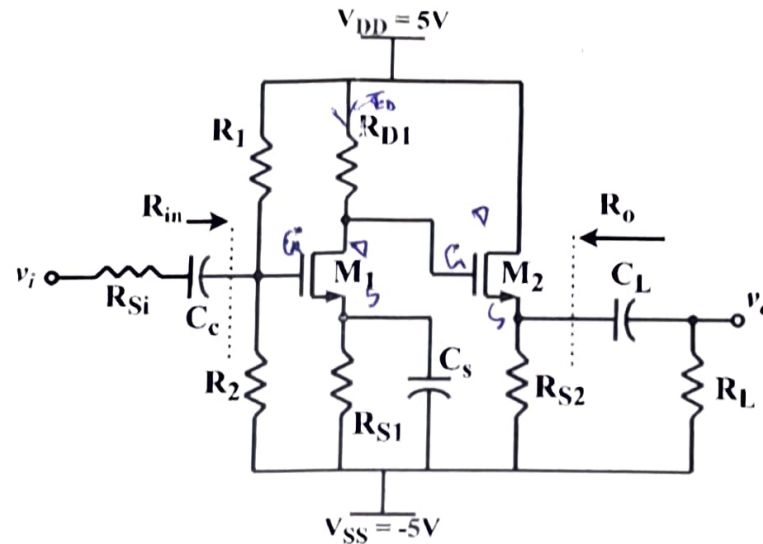


Figure 1

Calculate:

a. R_{S2}

(3 marks)

b. R_{D1}

(3 marks)

c. R_1 and R_2 assuming them to be equal

(3 marks)

d. Voltage gain ($A_v = v_o/v_i$) of the multistage amplifier, neglecting R_{Si} and R_L .

(4 marks)

IIT Roorkee: End Term Exam, Jan - May 2025

ECC - 204 Analog Circuits

Time : 3 hours

Maximum marks : 80

IMPORTANT INFORMATION

Use the following parameters for all the problems unless explicitly stated otherwise. $V_{DD} = 1.8V$, $V_{THN} = |V_{THP}| = 0.5V$. $\mu_n C_{ox} = 200 \mu A/V^2$, $\mu_p C_{ox} = 100 \mu A/V^2$ and $C_{ox} = 5fF/\mu m^2$. $\lambda = 1/50 V^{-1}$ for all transistors.

QUESTION - 1 (30 MARKS)

The inverting amplifier of Fig. 1(a) uses the two-stage differential amplifier of Fig. 1(b). V_1 is biased at $0.9V$. $V_2(t) = 0.9V + v_S(t)$, where $v_S(t)$ is the small signal which needs to be amplified. The required small signal gain from v_S to V_{out} is -2 .

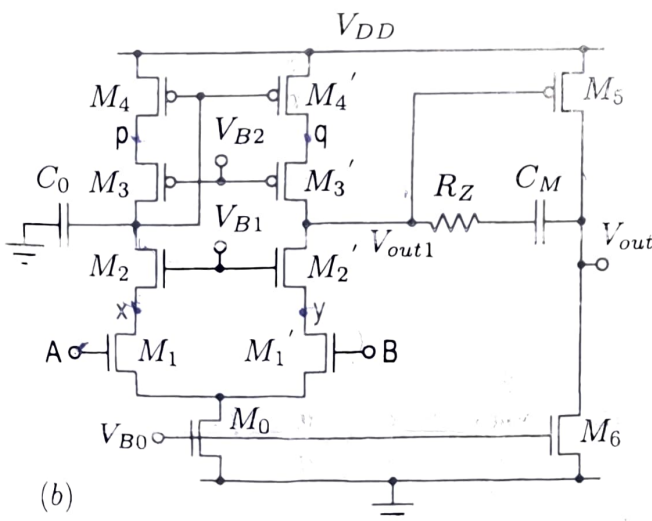
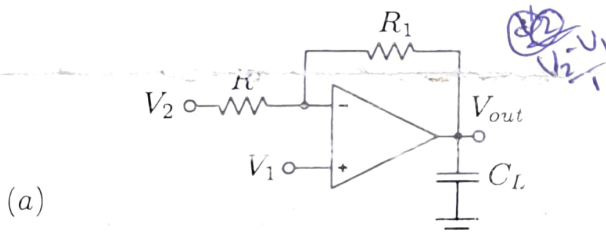


Fig. 1. Figure for question - 1

Assume C_{gs} , C_{gd} , C_{db} and C_{sb} are much smaller than C_0 , C_M and C_L in your analysis. You can

also ignore the poles/zeros which may be present at nodes p, q, x and y for simplicity. Design the differential amplifier (of Fig. 1(b)) for the following specifications.

- Peak-to-peak output voltage swing = $1.6V$
- $C_L = 1 pF$.
- Differential amplifier dc gain $\geq 80 dB$
- Unity gain bandwidth (UGB) = $100 MHz$.
- $R = 20 k\Omega$.
- Constant overdrive voltage for all the transistors. Make a choice such that all the transistors of Fig. 1(b) are in saturation and also that the device sizes (W/L) is minimized. Once you pick a choice of overdrive voltage, please use the same for all the transistors.
- The bias current in the second stage must be $3X$ the maximum low frequency output current to be sourced/sunk by the differential amplifier.
- The transconductance of M_5 , g_{m5} is three times that of the transconductance of M_1 , g_{m1} .

Please answer the following questions, which will help you in this design.

- Calculate the value of R_1 and the peak low frequency current to be sourced/sunk by the differential amplifier. (2 marks)
- Calculate the expression and value of dc gain of the two stage differential amplifier of Fig. 1(b). (2 marks)
- Calculate the expressions for the three pole locations of the differential amplifier of Fig. 1(b). Note that C_L needs to be considered at the output for these calculations. Also, ignore any parasitic poles due to R_Z . (6 marks)
- Calculate expressions for the two zero locations of the two stage differential amplifier of Fig. 1(b). State whether they are LHP or RHP zeros. (4 marks)
- How will you choose R_Z such a problematic zero is pushed to infinity? (1 mark)
- What are the values of C_M and the phase

Handwritten notes and calculations:

- $2 \times (g_{m2})$
- (R_1)
- $160 \times g_{m2} \times 1$
- $g_{m2} \times V$
- $160 \times g_{m2} \times 1$

margin of the differential amplifier? (4 marks).
[Hint: Assume that the pole due to C_0 exactly cancels the zero due to C_0 , to simplify the transfer function.]

- What are the bias currents in M_1 and M_6 ? (2 marks)
- What are the optimum values of the bias voltages V_{B0} , V_{B1} , V_{B2} so that the voltage headroom at V_{out1} is maximized. (3 marks)
- What are the (W/L) ratios of all the transistors M_1 to M_6 ? (6 marks)

QUESTION - 2 (15 MARKS)

Consider the common source amplifier of Fig. 2. V_{in} is the signal input of the amplifier. The resistor generates thermal noise which has a voltage power spectral density (PSD) of $4kTR$. The MOS transistors generate both thermal and flicker noise. The thermal noise current PSD of transistor M_x is $4kT\gamma g_{m_x}$, while its flicker noise voltage PSD is $K/(C_{ox}WLf)$. γ and K are constants. Ignore parasitic capacitors for your calculations.

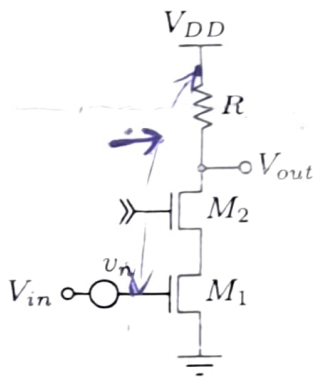


Fig. 2. Figure for question - 2

The total noise of the amplifier is represented by an equivalent *input-referred* noise source v_n as shown in Fig. 2. Calculate the expression for the power spectral density of v_n . (13 marks). From your result, can you comment on the noise contributions of M_1 and M_2 to the overall noise of the amplifier? (2 marks).

QUESTION-3 (20 MARKS)

Consider the circuit of Fig. 3. Assume the widths of M_3 and M_4 to be the same. The width of M_2 is four times that of M_1 . All transistors have the same length and also have $\lambda = 0$. Answer the following questions.

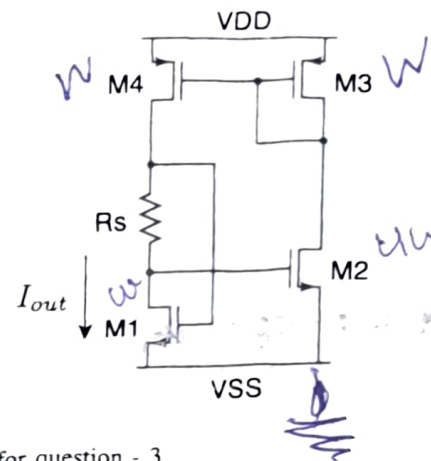


Fig. 3. Figure for question - 3

- Calculate the *quiescent* current I_{out} in terms of the resistance R_s and the transistor parameters - threshold voltage V_{THN} , mobility μ_n , dimensions W/L and oxide capacitance C_{ox} . How does I_{out} vary with temperature, assuming R_s is a constant? (10 marks).
- Calculate the small signal transconductance (g_{m1}) of M_1 using the quiescent I_{out} calculated earlier. How does g_{m1} vary with temperature, assuming R_s is a constant? (10 marks).

QUESTION-4 (15 MARKS)

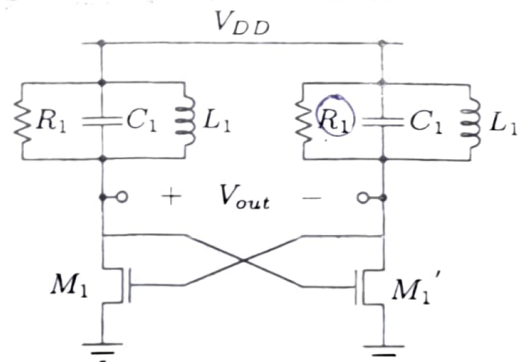


Fig. 4. Figure for question - 4

Consider the circuit of Fig. 4. Ignore channel length modulation in M_1 and M_1' . Answer the following questions.

- What is the functionality of the circuit? (2 marks).
- How do you choose the value of the transconductance of M_1/M_1' such that the circuit achieves its functionality? (10 marks).
- What is the frequency of the output signal V_{out} ? (3 marks).

ALL THE BEST !!

Important Formulae :-

N-channel

① Mosfet in Saturation :- $I_{DS} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) [V_{GSn} - V_{Thn}]^2 (1 + \lambda_n V_{DSn})$

② N-channel MOSFET in linear region $\left\{ \begin{array}{l} \therefore I_{DS} = \mu_n C_{ox} \left(\frac{W}{L} \right) \left\{ (V_{GSn} - V_{Thn}) V_{DSn} - \frac{V_{DSn}^2}{2} \right\} \end{array} \right.$

③ P-channel MOSFET in Saturation $\left\{ \begin{array}{l} \therefore I_{DS} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right) [V_{SGp} - |V_{Thp}|]^2 (1 + \lambda_p V_{DSp}) \end{array} \right.$

④ p-channel MOSFET in linear/triode region $\left\{ \begin{array}{l} \therefore I_{DS} = \mu_p C_{ox} \left(\frac{W}{L} \right) \left\{ (V_{SGp} - |V_{Thp}|) V_{DSp} - \frac{V_{DSp}^2}{2} \right\} \end{array} \right.$

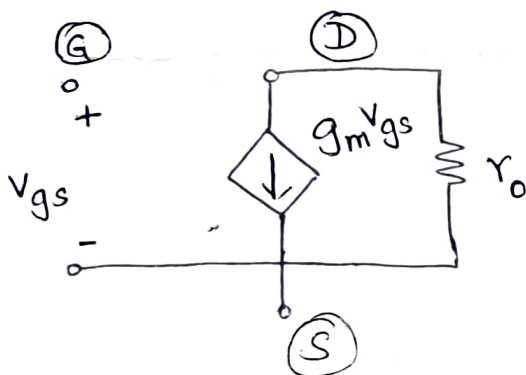
⑤ Expression for transconductance,

$$g_m = \frac{2 I_{DSQ}}{V_{GSQ} - V_{Th}} = \mu C_{ox} \left(\frac{W}{L} \right) (V_{GSQ} - V_{Th}) = \sqrt{2 \mu C_{ox} \frac{W}{L} I_{DSQ}}$$

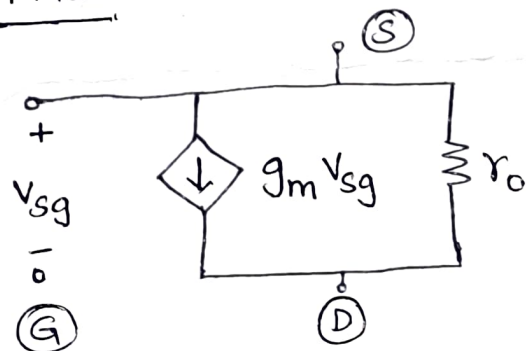
⑥ Expression for small signal output resistance $\left\{ \begin{array}{l} \text{(for transistor in saturation)} \end{array} \right. \quad r_{ds} = r_o = \frac{1}{\lambda I_{DSQ}}$

⑦ Small signal model :-

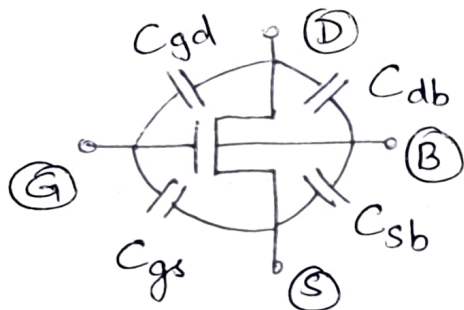
NMOS



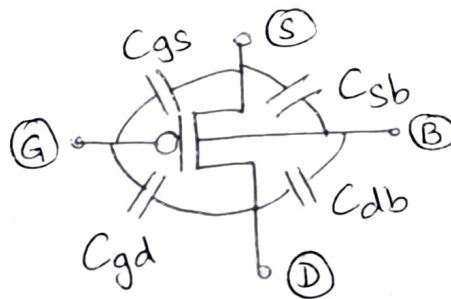
PMOS



⑧ MOSFET with parasitics

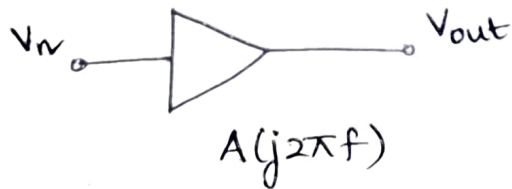


NMOS



PMOS

⑨ Calculation of o/p noise PSD, given input noise PSD



$$S_{V_{out}}(f) = |A(j2\pi f)|^2 S_{V_n}(f)$$

PSD of V_n

$S_{V_n}(f)$

ECC – 204: Analog Circuits (Part – 2) Quiz – 1 (Set - B)

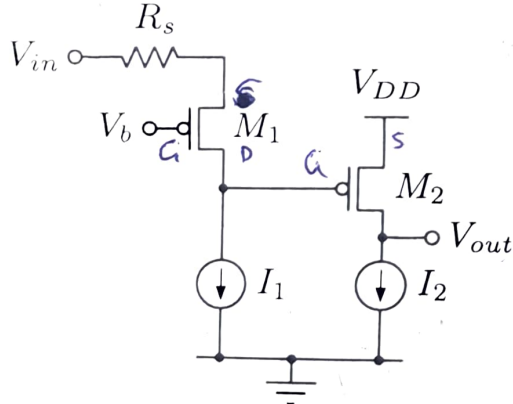
Maximum marks – 20

Duration – 30 minutes

Name:

Roll Number:

Question – 1:



Calculate the low frequency gain, pole/zero locations and hence, the overall transfer function from input V_{in} to output V_{out} . As discussed in class, each transistor has four parasitic capacitances – C_{gs} , C_{gd} , C_{sb} , C_{db} . V_b is DC bias voltage.